

V5 scientific integrity and reviewer implementation audit

Evidence boundary

Frozen source commit: `d5fefaa0cf7a31cf7d29830ed5b070e1d2dac3a2`. Current physical arrays, held-event predictions and training labels were not altered. MIKE rainfall/terrain/reference arrays and the public LarNO checkpoint are upstream materials, clearly attributed; A/B/C coupled simulations, fitted residual estimators and their diagnostics are this project's calculations. The manuscript does not claim all data are original observations.

Verified findings

The raw rainfall sum and active-cell mean use different masks. Geometric slope inspection finds no zero/negative conduit slope and 626 values near the design floor. No outfall has its own coordinate record. Spatial maps mark connecting terminals only. Peak-map differences are not instantaneous residuals. The held-event network increment is 0.047773 mm, event SD 0.037333 mm and sampling-seed macro SD 0.001819 mm. All are computed, not borrowed from the LarNO paper.

Implementation status

Review item	Status	Evidence or limitation
Rainfall mask and volume	Recomputed	576 intervals; verified numerical redistribution, not surveyed roof mask
Peak-map meaning	Corrected	$\max(C) - \max(B)$ explicitly separated from $C(t) - B(t)$
Slope zeros and outfalls	Recomputed	0 nonpositive; 626 near floor; outfall coordinates absent
Event and seed uncertainty	Recomputed	40 event-seed gains; event bootstrap conditional on exchangeability
Inference perturbations	Reframed	OOD sensitivity separated from retrained feature benefit
Native exchange formula/sign	Unit check only	Prescribed-head test double; no routed pipe benchmark
Full joint internal-step water ledger	Pending	30-minute cumulative records and final report only
Broader paired events / weather independence	Pending	Nine readable cases unpaired; dates/families unverified
Multi-event rainfall treatment / network capacities / tailwater	Pending	Only existing single-event setting sensitivities completed
Coarse prior / shifted rain / cross-scenario retraining	Pending	No new predictive claims
Unclamped depth error changes and extreme LarNO tails	Partially addressed	LarNO full-range maps and extremes rebuilt; clamp counts retained; full raw-depth change diagnostics still need inference replay
Authors and redistribution rights	Author action	Do not invent author declarations or upstream permission

Exact numerical evidence

File	SHA256
extended_study/output/reviewer_major_revision_v5/diagnostics/event_seed_gain.csv	38a470bd1594e725bdce7cfa3799018a417986646b2ea70a983ac0e83a6397a0
extended_study/output/reviewer_major_revision_v5/diagnostics/hydrograph_diagnostics.csv	d650c81cee3961e409672859b3a2cac4e95139ed320ab504b7763b93a59cf2c8
extended_study/output/reviewer_major_revision_v5/diagnostics/larno_extremes.json	1593c5d449e64c7b9bec5fbad13e9310b1bd3a1a4eb1f1796bb2214d3e482428
extended_study/output/reviewer_major_revision_v5/diagnostics/map_display_audit.csv	8d9d7b3d1190d4f5b1727e4274f56af29a4800472f184104d406a709a2d80ebe
extended_study/output/reviewer_major_revision_v5/diagnostics/native_exchange_unit_check.json	ef1f6529a83bec2b1b036952b12c66d62c718779d178d379db5a21281eb73f5c
extended_study/output/reviewer_major_revision_v5/diagnostics/network_geometry.json	4f694b48565d4e5f3a045afdfc8650d5ffe4faa85cca9756dbbfde62e0f46f8
extended_study/output/reviewer_major_revision_v5/diagnostics/rainfall_event.csv	9fafcaf913163ff5342fe24622daec99a62dbba229939cce0b828921af6cec5
extended_study/output/reviewer_major_revision_v5/diagnostics/rainfall_steps.csv	927a98aa697bf07c0d02e5d4f9aaa008c4e20f2a4b00c5a616ff62eca01f00bf
extended_study/output/reviewer_major_revision_v5/diagnostics/signed_residual_diagnostics.csv	e17a7c27b821c13d73f4ac5cc8a288e1ef2f6fb38242c703cflc7d649ef9af50
extended_study/output/reviewer_major_revision_v5/diagnostics/summary.json	3953ac835be6b93713e6efc661dbcea8432a2f7794c452848a3b17b40fa6f711
extended_study/output/reviewer_major_revision_v5/diagnostics/water_ledger_saved_times.csv	f694e6605b502d64038af47257aea863c4637c10e8d388fed95dd9176598a6b6

Reproduction commands

```
python extended_study/audit_reviewer_v5_evidence.py
python extended_study/test_v5_native_exchange.py
python extended_study/generate_reviewer_v5_figures.py
python extended_study/generate_reviewer_v5_documents.py
python extended_study/render_reviewer_v5_documents.py
```

Primary training remains `extended_study/run_final_hybrid_major_revision.py` (fixed 140 iterations, no residual clipping). Historical V3 tuning is not a reproduction command for the V4/V5 primary results. Figures 1/2/6/7/9/10/11 and supplementary maps are rebuilt by the V5 figure script; unchanged physical-quality, hydrograph and peak-map illustrations originate from the frozen V4 package. Figure 5's data formula is audited in `plot_reviewer_v3_physics.py`. Full CSV evidence remains alongside concise manuscript tables; removing duplicate plots does not remove inconvenient results.

Writing sources and source verification

The revision follows concrete, evidence-led prose and distinguishes methods, results and interpretation. It does not manufacture an author voice, field observations or stronger conclusions to evade AI detection. Sources consulted: Manchester Academic Phrasebank (<https://www.phrasebank.manchester.ac.uk/>), KKKKhazix/human-writing (<https://github.com/KKKKhazix/human-writing>), petergyang/no-ai-slop (<https://github.com/petergyang/no-ai-slop>). These are editorial guidance, not scientific evidence.

MIKE's coupled origin is established in Section 4.1 of the supplied Cao et al. paper, DOI 10.1016/j.jhydrol.2026.135686. The current public model card documents 13 released input channels and the 64/16 Futian split (https://github.com/holmescao/LarNO/blob/main/huggingface/model_card.md). Current Itzi stable documentation resolves to 26.6, while the frozen calculations use 25.4; the current documentation version must not be represented as the executed engine. Normal-depth outfalls are conceptual hydraulic boundary conditions, not measurements of receiving-water levels.

Submission readiness

This revision improves presentation and evidence transparency. It does not close the pending physical validation and scenario-generalisation requests. Numerical consistency checks alone are not grounds to declare the study calibrated or ready for unconditional submission.